

"" . img_tag(ASSETS_LOGO, 'Logo Perpustakaan Daerah', 'width:300px;') . ""

DOKUMENTASI SISTEM

INFORMASI

PERPUSTAKAAN DAERAH

Versi 1.0

Sistem Informasi Manajemen Perpustakaan Daerah VERSI 1

PHP Native + MySQL/MariaDB

Dokumen Teknis Lengkap

ERD, Diagram, Algoritma, Keamanan, Deployment

Disusun Oleh:

Tim Pengembang Sistem

Tanggal: "" . PDF_DATE . ""

Versi: "" . PDF_VERSION . "" (V1)

Terakhir Diperbarui: "" . PDF_DATE . ""

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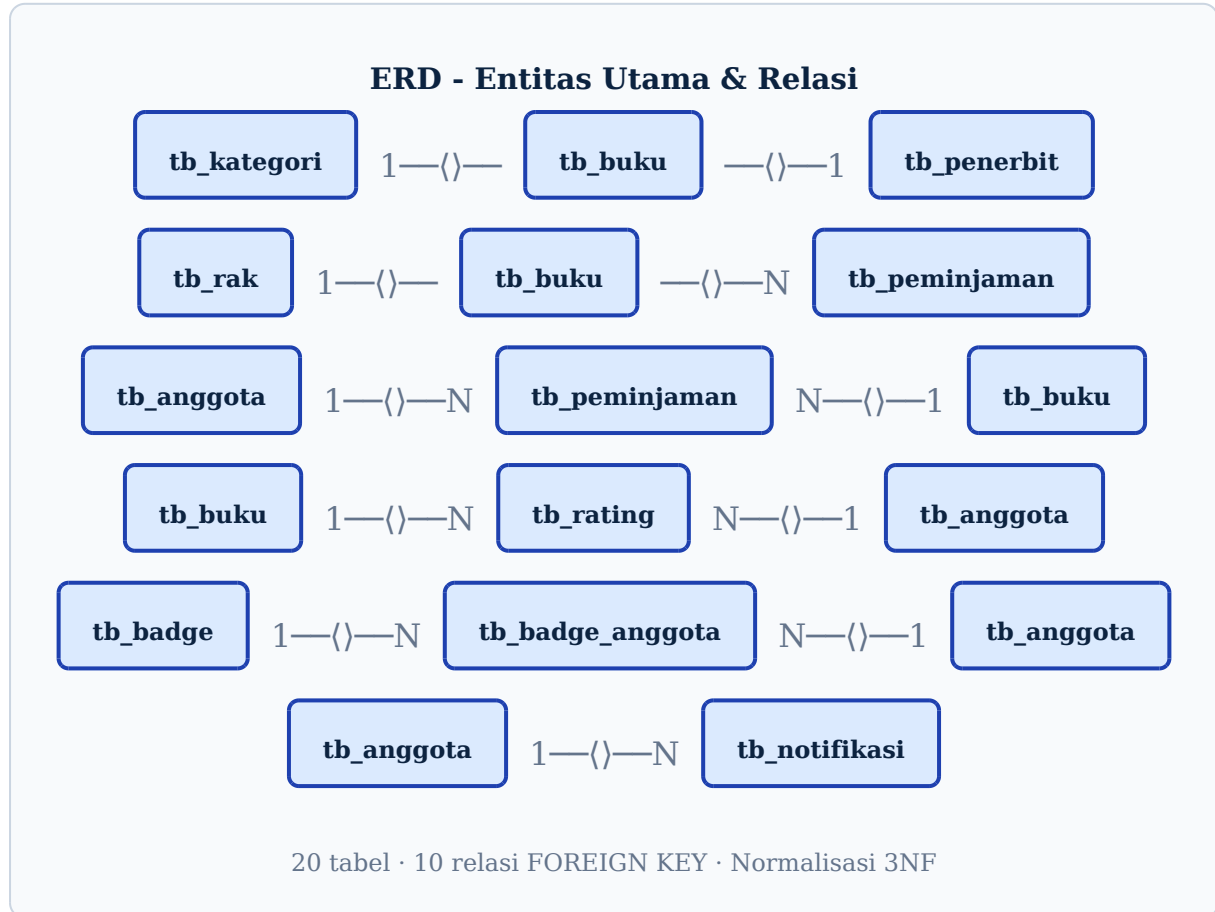
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1. Entity Relationship Diagram (ERD)

1.1 Diagram ER



1.2 Daftar Tabel Database

No	Nama Tabel	Deskripsi	Record
1	tb_anggota	Data anggota perpustakaan	10
2	tb_badge	Definisi badge pencapaian	8
3	tb_badge_anggota	Badge yang diraih anggota	6
4	tb_banner	Banner CMS	0
5	tb_berita	Berita CMS	0

6	tb_buku	Katalog buku	11
7	tb_faq	FAQ	4
8	tb_jam_buka	Jam operasional	7
9	tb_kategori	Kategori buku	8
10	tb_layanan	Layanan perpustakaan	6
11	tb_notifikasi	Sistem notifikasi	35
12	tb_peminjaman	Transaksi peminjaman	27
13	tb_penerbit	Data penerbit	7
14	tb_pengaturan	Pengaturan sistem	7
15	tb_pengumuman	Pengumuman CMS	1
16	tb_permission	Hak akses role	30
17	tb_rak	Lokasi rak buku	5
18	tb_rating	Rating dan komentar buku	1
19	tb_sesi_login	Sesi login persisten	1
20	tb_user	Data user sistem (admin/petugas)	2

1.3 Tabel Inti

tb_anggota (Data Anggota)

Kolom	Tipe	Constraint	Keterangan
id_anggota	INT	PK, AUTO_INCREMENT	ID unik anggota
nama_lengkap	VARCHAR(100)	NOT NULL	Nama lengkap
email	VARCHAR(100)	UNIQUE, NOT NULL	Email (login)

password	VARCHAR(255)	NOT NULL	Bcrypt hash
no_hp	VARCHAR(15)	NULLABLE	Nomor HP
alamat	TEXT	NULLABLE	Alamat lengkap
foto	VARCHAR(255)	NULLABLE	Path foto profil
status	ENUM	DEFAULT 'aktif'	aktif/nonaktif/dibanned
created_at	TIMESTAMP	DEFAULT NOW()	Waktu registrasi

tb_buku (Katalog Buku)

Kolom	Tipe	Constraint	Keterangan
id_buku	INT	PK, AUTO_INCREMENT	ID unik buku
isbn	VARCHAR(20)	UNIQUE, NULLABLE	ISBN buku
judul	VARCHAR(200)	NOT NULL	Judul buku
pengarang	VARCHAR(100)	NOT NULL	Nama pengarang
id_kategori	INT	FK → tb_kategori	ID kategori
id_penerbit	INT	FK → tb_penerbit	ID penerbit
id_rak	INT	FK → tb_rak	ID lokasi rak
tahun_terbit	YEAR	NULLABLE	Tahun terbit
jumlah_buku	INT	DEFAULT 0	Total stok
stok_buku	INT	DEFAULT 0	Stok tersedia
cover	VARCHAR(255)	NULLABLE	Path cover buku
sinopsis	TEXT	NULLABLE	Singkatan cerita

tb_peminjaman (Transaksi)

Kolom	Tipe	Constraint	Keterangan
-------	------	------------	------------

id_peminjaman	INT	PK, AUTO_INCREMENT	ID transaksi
id_anggota	INT	FK → tb_anggota	ID peminjam
id_buku	INT	FK → tb_buku	ID buku
kode_peminjaman	VARCHAR(20)	UNIQUE	Kode unik PMJ-YYYYMMDD-XXX
status	ENUM	DEFAULT 'dipinjam'	dipinjam/terlambat/dikembalikan/menuju
tgl_pinjam	DATE	NOT NULL	Tanggal pinjam
tgl_harus_kembali	DATE	NOT NULL	Batas kembali
tgl_kembali	DATE	NULLABLE	Tanggal aktual kembali
denda	INT	DEFAULT 0	Denda (Rp)
kondisi	VARCHAR(50)	NULLABLE	Kondisi buku saat kembali

1.4 Relationship Summary

Parent	Child	Foreign Key	Kardinalitas
tb_kategori	tb_buku	id_kategori	1:N
tb_penerbit	tb_buku	id_penerbit	1:N
tb_rak	tb_buku	id_rak	1:N
tb_anggota	tb_peminjaman	id_anggota	1:N
tb_buku	tb_peminjaman	id_buku	1:N
tb_buku	tb_rating	id_buku	1:N
tb_anggota	tb_rating	id_anggota	1:N
tb_badge	tb_badge_anggota	id_badge	1:N
tb_anggota	tb_badge_anggota	id_anggota	1:N

tb_anggota

tb_notifikasi

id_anggota

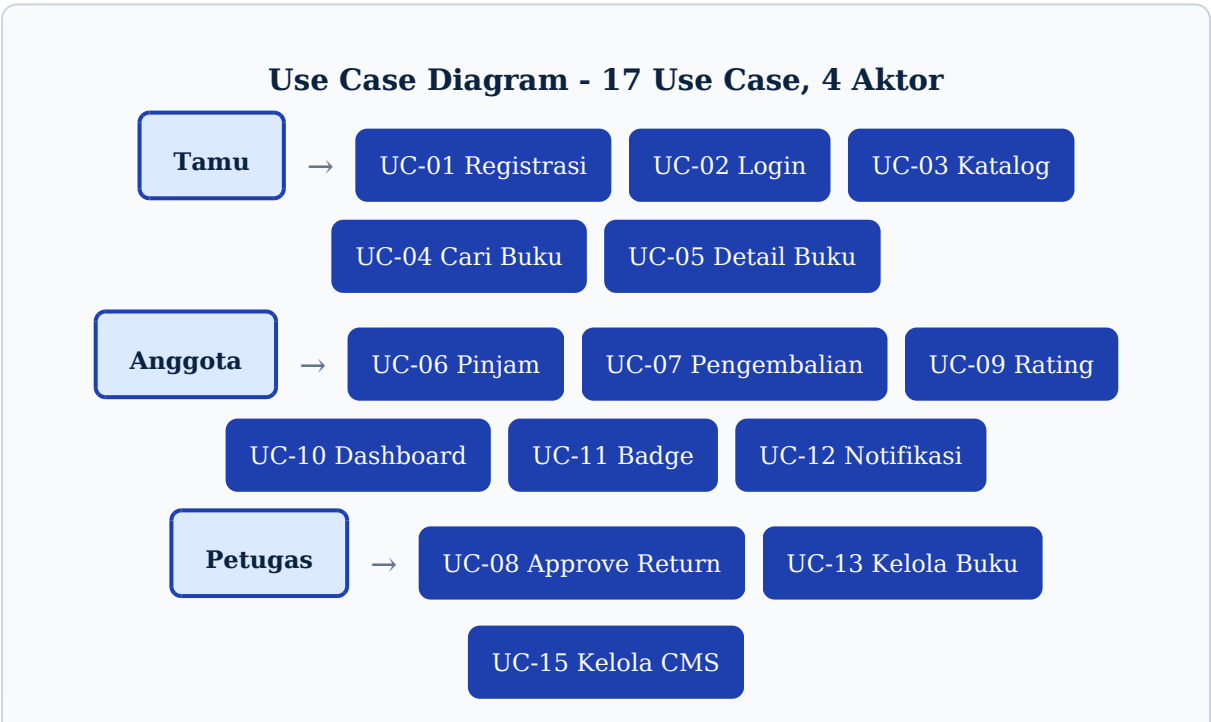
1:N

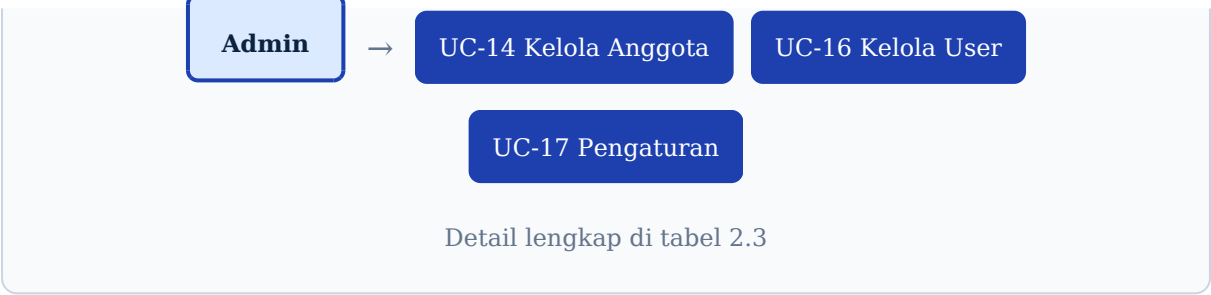
2. Diagram Use Case

2.1 Aktor Sistem

Aktor	Deskripsi	Level Akses
Tamu (Guest)	Pengunjung yang belum login	Baca publik saja
Anggota	Anggota terdaftar yang sudah login	Buku, Peminjaman, Profil, Rating
Petugas	Staf perpustakaan	Kelola konten + approve pengembalian
Admin	Administrator sistem	Akses penuh

2.2 Use Case Diagram





2.3 Use Case Details

No	Use Case	Aktor	Deskripsi
UC-01	Registrasi Akun	Tamu	Membuat akun anggota baru
UC-02	Login	Semua	Masuk ke sistem
UC-03	Lihat Katalog	Semua	Menampilkan daftar buku
UC-04	Cari Buku	Semua	Pencarian buku dengan filter
UC-05	Detail Buku	Semua	Menampilkan detail + rating
UC-06	Pinjam Buku	Anggota	Melakukan peminjaman
UC-07	Ajukan Pengembalian	Anggota	Mengajukan pengembalian buku
UC-08	Approve Pengembalian	Admin/Petugas	Menyetujui pengembalian
UC-09	Beri Rating	Anggota	Rating + komentar buku
UC-10	Kelola Dashboard	Anggota	Melihat statistik personal
UC-11	Lihat Badge	Anggota	Menampilkan pencapaian
UC-12	Lihat Notifikasi	Anggota	Panel notifikasi
UC-13	Kelola Buku	Admin/Petugas	CRUD data buku
UC-14	Kelola Anggota	Admin	CRUD data anggota
UC-15	Kelola CMS	Admin/Petugas	Berita, FAQ, banner, dll
UC-16	Kelola User	Admin	Manajemen user sistem

2.4 Permission Matrix

Modul	Tamu	Anggota	Petugas	Admin
Katalog Buku	☐ Lihat	☐ Lihat	☐ CRUD	☐ CRUD
Peminjaman	☐	☐ Pinjam	☐ Approve	☐ Full
Rating	☐	☐ Tulis	☐ Lihat	☐ CRUD
Anggota	☐	☐ Profil	☐ Lihat	☐ CRUD
CMS	☐ Lihat	☐ Lihat	☐ Tulis	☐ CRUD
User	☐	☐	☐	☐ CRUD
Pengaturan	☐	☐	☐	☐ CRUD

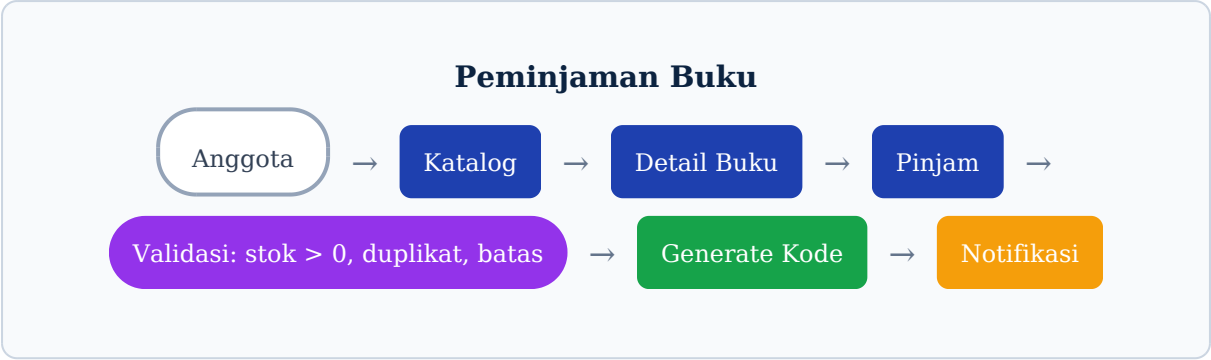
3. User Flow (Alur Pengguna)

3.1 Flow Registrasi



Langkah	Aksi	Validasi
1	Buka form registrasi	Honeypot field
2	Isi: nama, email, password, no HP, alamat	Required fields
3	Submit form	Email unik, password $\geq$ 6 char
4	Insert ke tb_anggota	Bcrypt hash password
5	Redirect ke login	Flash message sukses

3.2 Flow Peminjaman



Langkah	Aksi	Validasi
1	Browse katalog	-
2	Pilih buku → Detail	-
3	Klik "Pinjam Buku Ini"	Stok > 0, belum dipinjam, batas ≤ 2
4	Konfirmasi pinjam	Generate kode PMJ-YYYYMMDD-XXX
5	Update stok -1	Transaction rollback jika gagal
6	Notifikasi ke admin	Tipe: borrowing

3.3 Flow Pengembalian



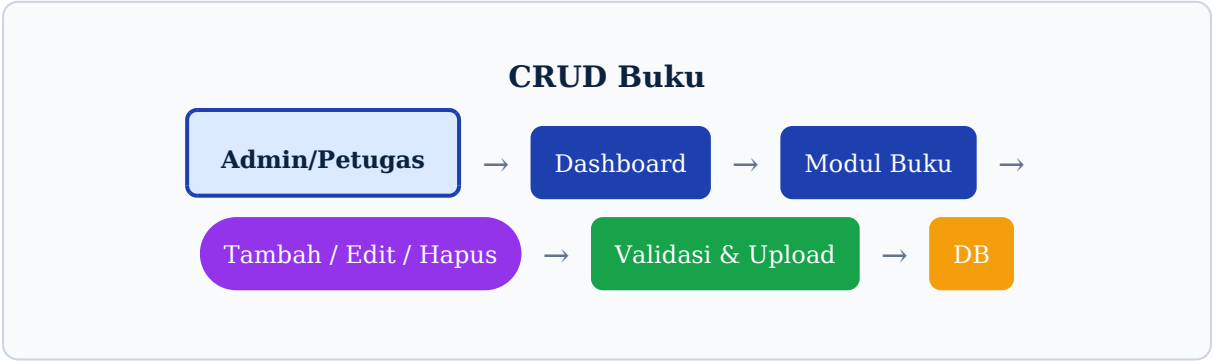
Langkah	Aksi	Validasi
1	Buka pengembalian.php	Tampilkan buku dipinjam/terlambat
2	Klik "Ajukan Pengembalian"	Catatan kondisi buku
3	Status → menunggu_pengembalian	Notifikasi ke admin
4	Admin review & approve	Kondisi, hitung denda
5	Update stok +1	Transaction rollback jika gagal
6	Badge sync	Cek total_pinjam, tepat_waktu

3.4 Flow Login

Langkah	Aksi	Validasi
1	Buka login.php	-
2	Isi email + password	Rate limiting (5/15 min)
3	password_verify()	Bcrypt comparison
4	session_regenerate_id()	Prevent session fixation
5	Update last_login	Badge sync on login
6	Redirect ke akun.php	Flash message sukses

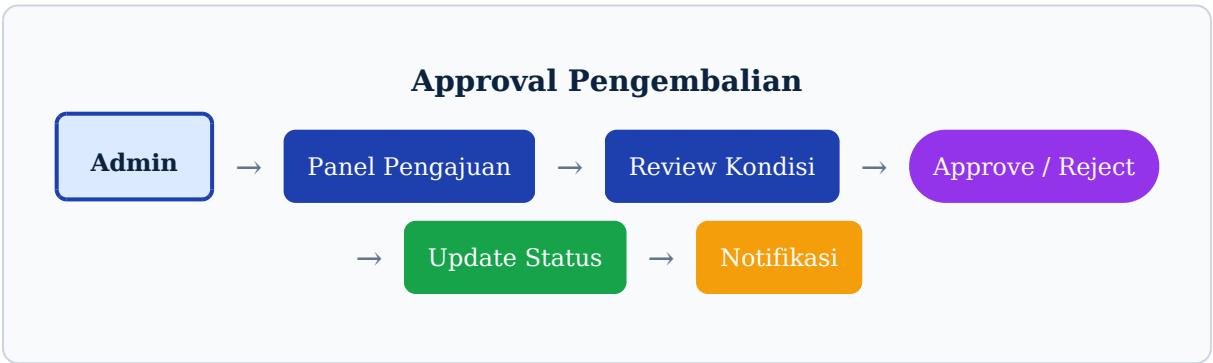
4. Admin Flow (Alur Administrator)

4.1 Flow Kelola Buku



Aksi	Input	Validasi	Output
Tambah Buku	Form lengkap + cover	Required: judul, pengarang, kategori, penerbit, rak	Insert + upload cover
Edit Buku	Form existing	ID valid + exist	Update record
Hapus Buku	ID buku	Tidak ada peminjaman aktif	Soft/Hard delete
Verifikasi Stok	ID buku	$\text{jumlah_buku} \geq \text{stok_buku}$	Trigger validasi

4.2 Flow Approve Pengembalian



Langkah	Aksi	Kalkulasi
---------	------	-----------

1	Lihat daftar pengajuan	Filter: menunggu_pengembalian
2	Review kondisi buku	Catatan dari anggota
3	Hitung denda	(hari terlambat × 10.000)
4	Approve	Update status, stok +1
5	Notifikasi ke anggota	Tipe: return_approved

4.3 Flow CMS

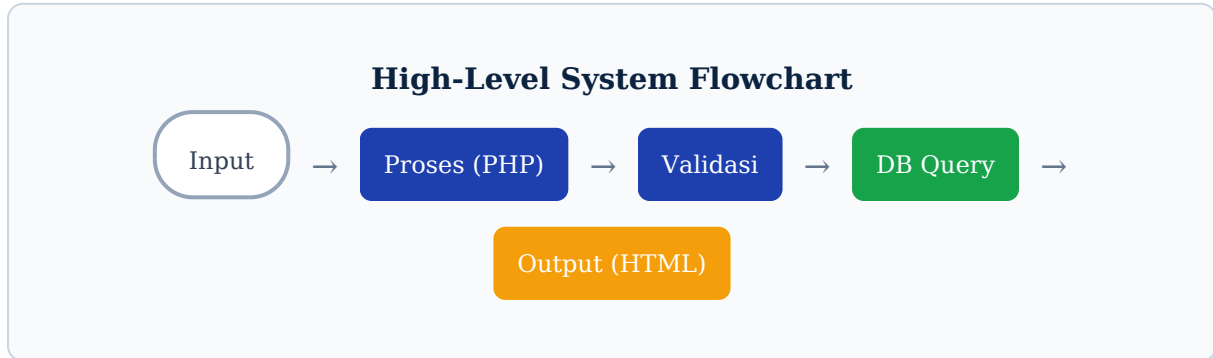
Modul	Aksi	Target
Berita	CRUD	tb_berita
Pengumuman	CRUD	tb_pengumuman
Banner	CRUD + Upload	tb_banner
Layanan	CRUD	tb_layanan
FAQ	CRUD	tb_faq
Jam Buka	CRUD	tb_jam_buka
Kontak	Update	tb_pengaturan

4.4 Access Control Matrix

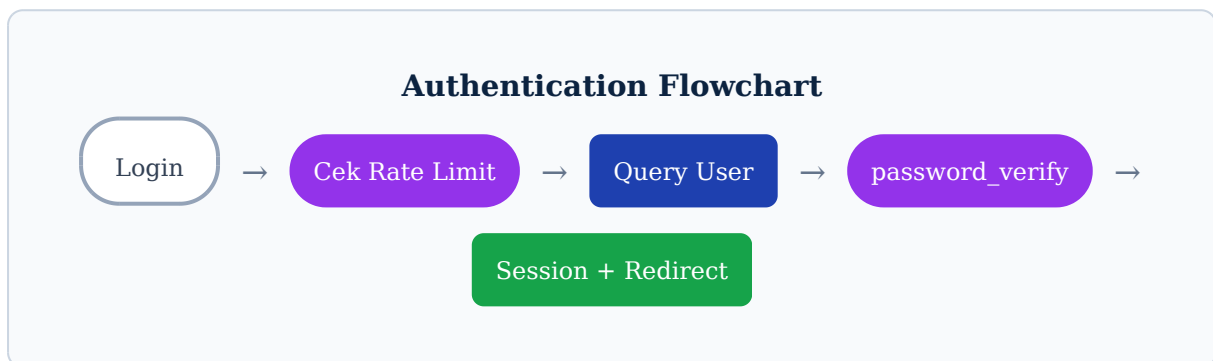
Halaman	Tamu	Anggota	Petugas	Admin
index.php	☐	☐	☐	☐
katalog_buku.php	☐	☐	☐	☐
login.php	☐	☐	☐	☐
akun.php	☐	☐	☐	☐
pinjam.php	☐	☐	☐	☐
pengembalian.php	☐	☐	☐	☐

5. Flowchart

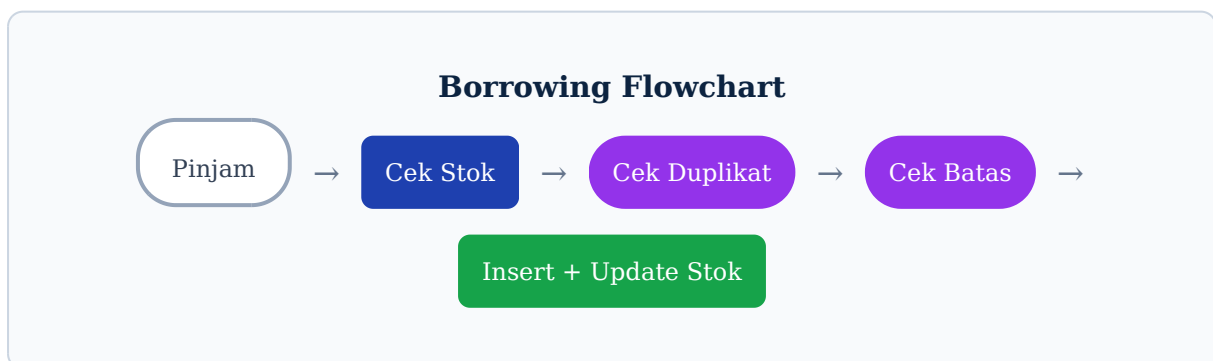
5.1 Flowchart Sistem Utama



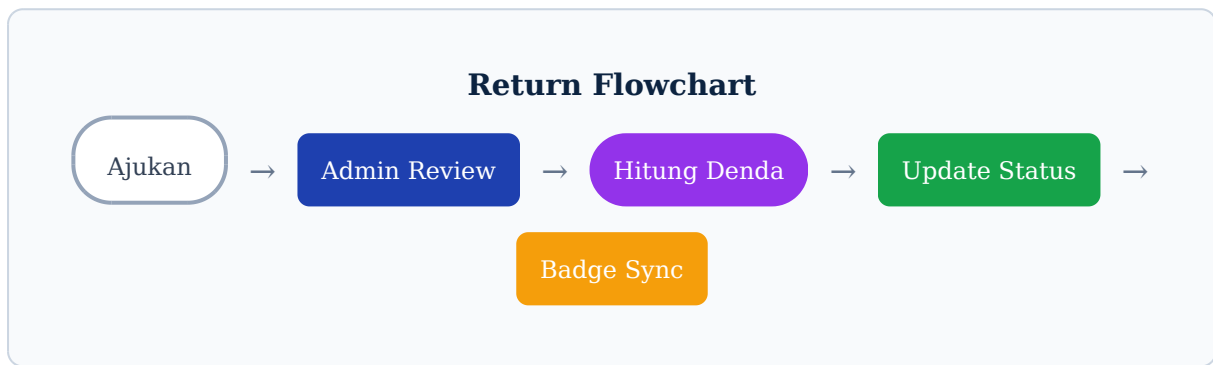
5.2 Flowchart Autentikasi



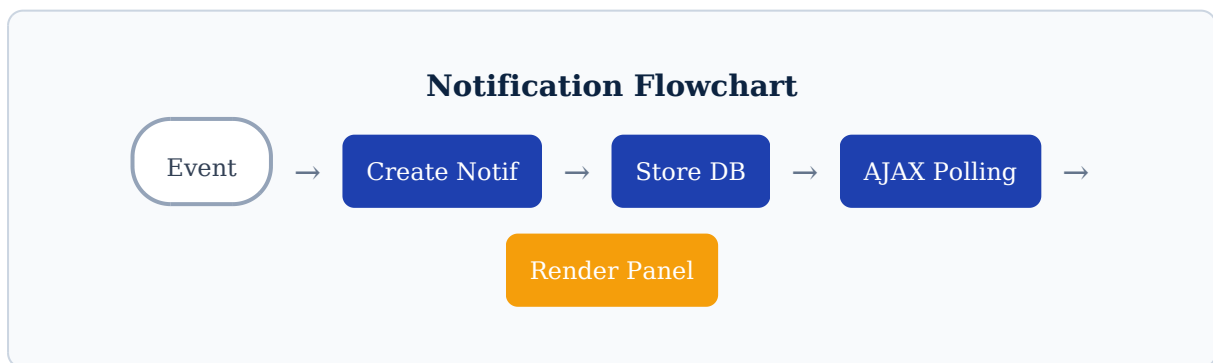
5.3 Flowchart Peminjaman



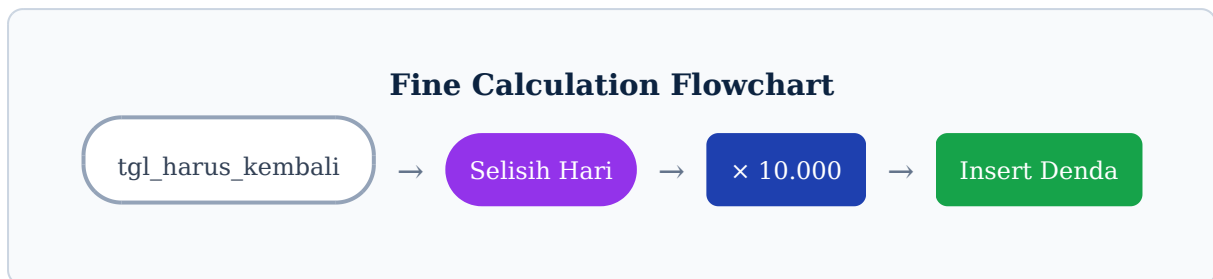
5.4 Flowchart Pengembalian



5.5 Flowchart Notifikasi



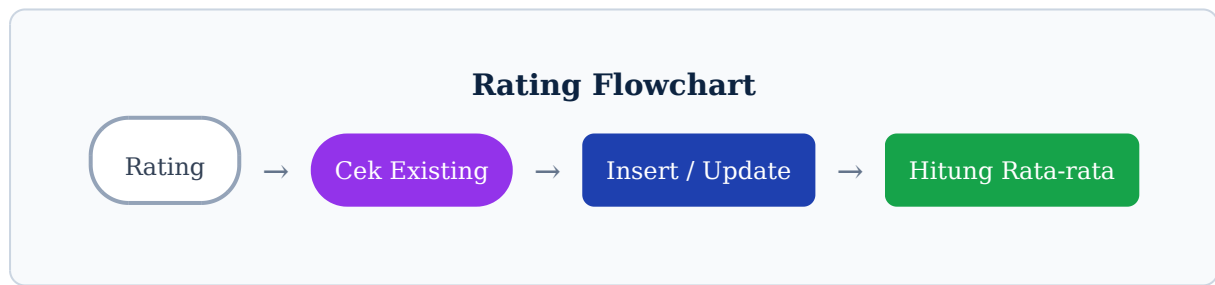
5.6 Flowchart Kalkulasi Denda



5.7 Flowchart Badge Sync



5.8 Flowchart Rating



6. Algoritma

6.1 Algoritma Login

```
FUNGSI login(email, password):
    CEK rate_limit(email)
    JIKA rate_limited MAKA
        TAMPILKAN "Akun terkunci 15 menit"
        KEMBALIKAN false
    AKHIR JIKA

    user = SELECT * FROM tb_anggota WHERE email = ?
    JIKA user tidak ditemukan MAKA
        INCREMENT login_attempts
        TAMPILKAN "Email tidak terdaftar"
        KEMBALIKAN false
    AKHIR JIKA

    JIKA user.status != 'aktif' MAKA
        TAMPILKAN "Akun nonaktif"
        KEMBALIKAN false
    AKHIR JIKA

    JIKA password_verify(password, user.password) MAKA
        session_regenerate_id(true)
        UPDATE last_login = NOW()
        RESET login_attempts
        badge_sync(user.id_anggota)
        KEMBALIKAN true
    AKHIR JIKA

    INCREMENT login_attempts
    TAMPILKAN "Password salah"
    KEMBALIKAN false
AKHIR FUNGSI
```

6.2 Algoritma Peminjaman

```

FUNGSI pinjam_buku(id_anggota, id_buku):
    buku = SELECT * FROM tb_buku WHERE id_buku = ?

    JIKA buku.stok_buku <= 0 MAKA
        KEMBALIKAN ERROR "Stok habis"
    AKHIR JIKA

    existing = SELECT COUNT(*) FROM tb_peminjaman
        WHERE id_anggota = ? AND id_buku = ?
        AND status IN ('dipinjam', 'terlambat')
    JIKA existing > 0 MAKA
        KEMBALIKAN ERROR "Sudah dipinjam"
    AKHIR JIKA

    total = SELECT COUNT(*) FROM tb_peminjaman
        WHERE id_anggota = ?
        AND status IN ('dipinjam', 'terlambat', 'menunggu_pengembalian')
    JIKA total >= 2 MAKA
        KEMBALIKAN ERROR "Batas peminjaman tercapai"
    AKHIR JIKA

    kode = generate_kode() // PMJ-YYYYMMDD-XXX
    tgl_harus = tanggal_hari_ini + 7 hari

    BEGIN TRANSACTION
        INSERT INTO tb_peminjaman
        UPDATE tb_buku SET stok_buku = stok_buku - 1
    COMMIT

    notif_create(id_anggota, 'borrowing')
    KEMBALIKAN kode
AKHIR FUNGSI

```

6.3 Algoritma Pengembalian

```

FUNGSI ajukan_pengembalian(id_peminjaman, catatan):
    UPDATE tb_peminjaman SET
        status = 'menunggu_pengembalian',
        catatan = ?
    WHERE id_peminjaman = ?

```

```

    notif_create(id_admin, 'return_request')
    KEMBALIKAN true
AKHIR FUNGSI

FUNGSI approve_pengembalian(id_peminjaman, kondisi):
    peminjaman = SELECT * FROM tb_peminjaman WHERE id_peminjaman = ?
    hari_terlambat = MAX(0, hari_sekarang - tgl_harus_kembali)
    denda = hari_terlambat × 10000

    BEGIN TRANSACTION
        UPDATE tb_peminjaman SET
            status = 'dikembalikan',
            tgl_kembali = hari_ini,
            denda = ?,
            kondisi = ?
        UPDATE tb_buku SET stok_buku = stok_buku + 1
    COMMIT

    notif_create(id_anggota, 'return_approved')
    badge_sync(id_anggota)
    KEMBALIKAN true
AKHIR FUNGSI

```

6.4 Algoritma Pencarian

```

FUNGSI cari_buku(kata_kunci, kategori_id, sort_by):
    query = "SELECT b.*, k.nama_kategori, p.nama_penerbit
        FROM tb_buku b
        LEFT JOIN tb_kategori k ON b.id_kategori = k.id_kategori
        LEFT JOIN tb_penerbit p ON b.id_penerbit = p.id_penerbit
        WHERE (b.judul LIKE ? OR b.pengarang LIKE ?)"

    params = ["%kata_kunci%", "%kata_kunci%"]

    JIKA kategori_id != 0 MAKA
        query += " AND b.id_kategori = ?"
        TAMBAH params kategori_id
    AKHIR JIKA

    QUERY query DENGAN params
    KEMBALIKAN hasil

```

AKHIR FUNGSI

6.5 Algoritma Kalkulasi Denda

```
FUNGSI hitung_denda(tgl_harus_kembali):  
    selisih = tgl_sekarang - tgl_harus_kembali  
    JIKA selisih <= 0 MAKA  
        KEMBALIKAN 0  
    AKHIR JIKA  
    denda = selisih × 10000  
    KEMBALIKAN denda  
AKHIR FUNGSI
```

6.6 Algoritma Badge Sync

```
FUNGSI badge_sync(id_anggota):  
    total_pinjam = COUNT(peminjaman SELESAI)  
    total_review = COUNT(rating DENGAN komentar)  
    tepat_waktu = COUNT(peminjaman TANPA denda > 0)  
  
    UNTUK SETIAP badge IN tb_badge:  
        JIKA badge.metrik == 'total_pinjam' MAKA  
            cek = total_pinjam >= badge.syarat  
        AKHIR JIKA  
        JIKA badge.metrik == 'total_review' MAKA  
            cek = total_review >= badge.syarat  
        AKHIR JIKA  
        JIKA badge.metrik == 'tepat_waktu' MAKA  
            cek = tepat_waktu >= badge.syarat  
        AKHIR JIKA  
  
        JIKA cek DAN belum_dimiliki MAKA  
            INSERT INTO tb_badge_anggota  
        AKHIR JIKA  
    AKHIR UNTUK  
AKHIR FUNGSI
```

6.7 Algoritma Notifikasi

```

FUNGSI notif_create(id_anggota, tipe, judul, pesan):
    INSERT INTO tb_notifikasi (
        id_anggota, tipe, judul, pesan,
        dibaca_at, created_at
    ) VALUES (?, ?, ?, ?, NULL, NOW())
    KEMBALIKAN true
AKHIR FUNGSI

```

```

FUNGSI notif_list(id_anggota, limit):
    unread = COUNT(*) WHERE dibaca_at IS NULL
    items = SELECT * FROM tb_notifikasi
        WHERE id_anggota = ?
        ORDER BY created_at DESC
        LIMIT ?
    KEMBALIKAN {unread, items}
AKHIR FUNGSI

```

6.8 Algoritma Rating

```

FUNGSI rate_buku(id_buku, id_anggota, rating, komentar):
    existing = SELECT * FROM tb_rating
        WHERE id_buku = ? AND id_anggota = ?

    JIKA existing MAKA
        UPDATE tb_rating SET rating = ?, komentar = ?
    AKHIR JIKA

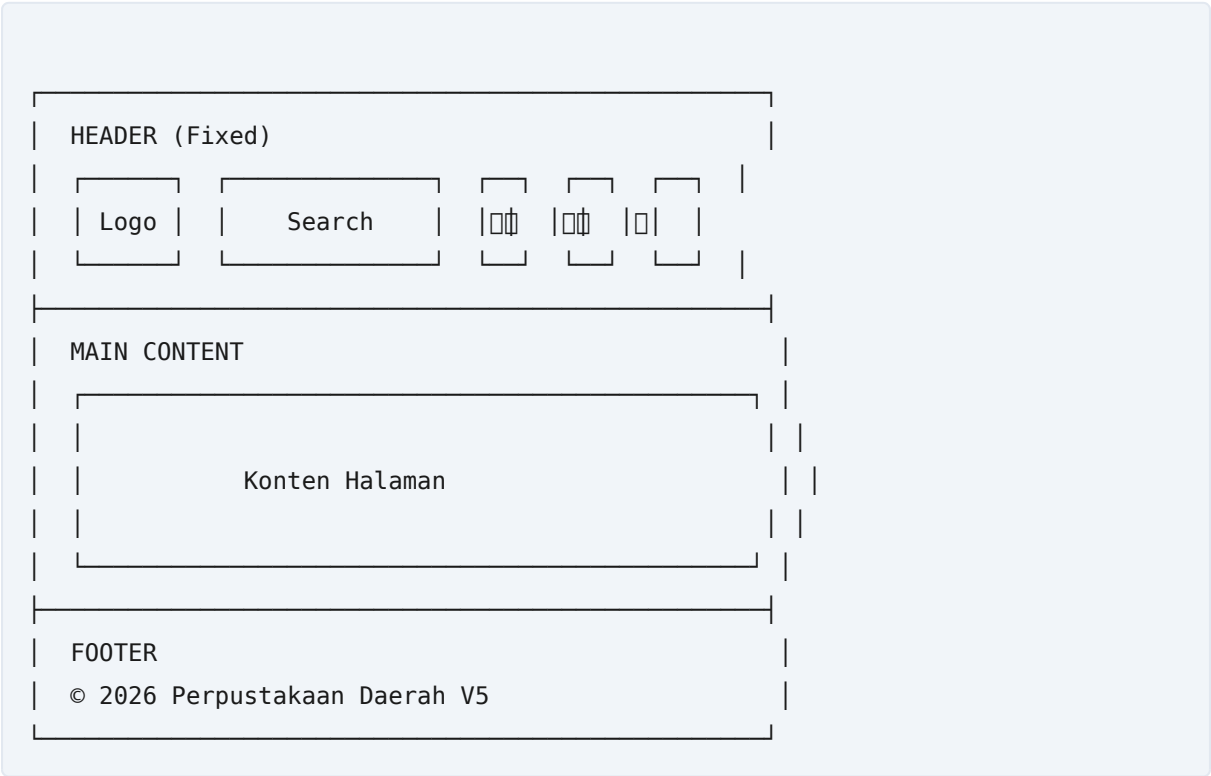
    JIKA bukan existing MAKA
        INSERT INTO tb_rating
    AKHIR JIKA

    rata2 = AVG(rating) WHERE id_buku = ?
    UPDATE tb_buku SET rata_rating = ?
    KEMBALIKAN true
AKHIR FUNGSI

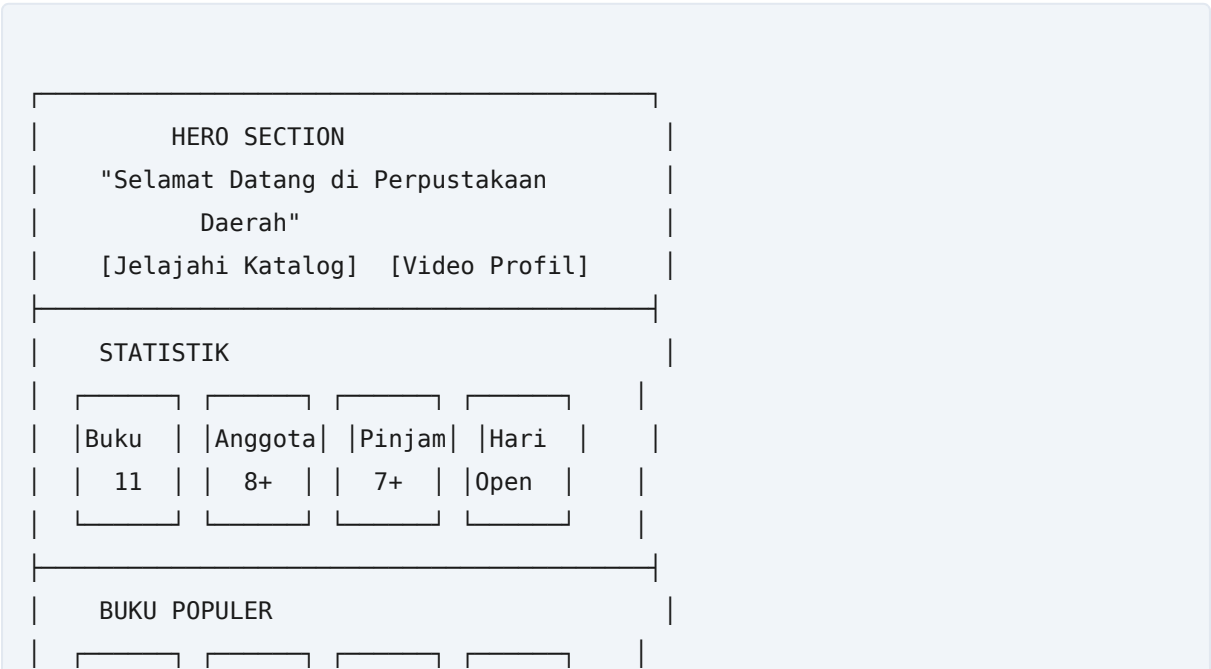
```

7. Mockup UI

7.1 Layout Template



7.2 Landing Page



Cover	Cover	Cover	Cover
Buku 1	Buku 2	Buku 3	Buku 4
★ 4.5	★ 4.0	★ 5.0	★ 4.0

7.3 Katalog Buku

PENCARIAN & FILTER			
Cari buku...		Filter ▼	
DAFTAR BUKU			
Cover	Cover	Cover	Cover
Buku 1	Buku 2	Buku 3	Buku 4
Judul	Judul	Judul	Judul
Kateg.	Kateg.	Kateg.	Kateg.
Stok 5	Stok 8	Stok 3	Stok 6
◀ 1 2 3 4 5 ▶			

7.4 Detail Buku

Cover Buku ★★★★☆	JUDUL BUKU
	Pengarang: ...
	Penerbit: ...
	Kategori: ...
	Tahun: 2024
	Stok: 5 buku
	ISBN: ...
PINJAM BUKU INI	

SINOPSIS

Lorem ipsum dolor sit amet...

RATING & KOMENTAR

★★★★	"Buku bagus" - Ahmad
★★★★☆	"Cukup menarik" - Budi

7.5 Form Peminjaman

KONFIRMASI PEMINJAMAN

Judul: Laskar Pelangi

Pengarang: Andrea Hirata

Tanggal Pinjam: 02/09/2026

Tanggal Kembali: 09/09/2026

Denda Keterlambatan: Rp 10.000/hari

Batal

Konfirmasi

7.6 Dashboard Anggota

SELAMAT DATANG, AHMAD!		
STATISTIK		
Total	Sedang	Denda
Pinjam: 5	Dipinjam:2	Rp 0

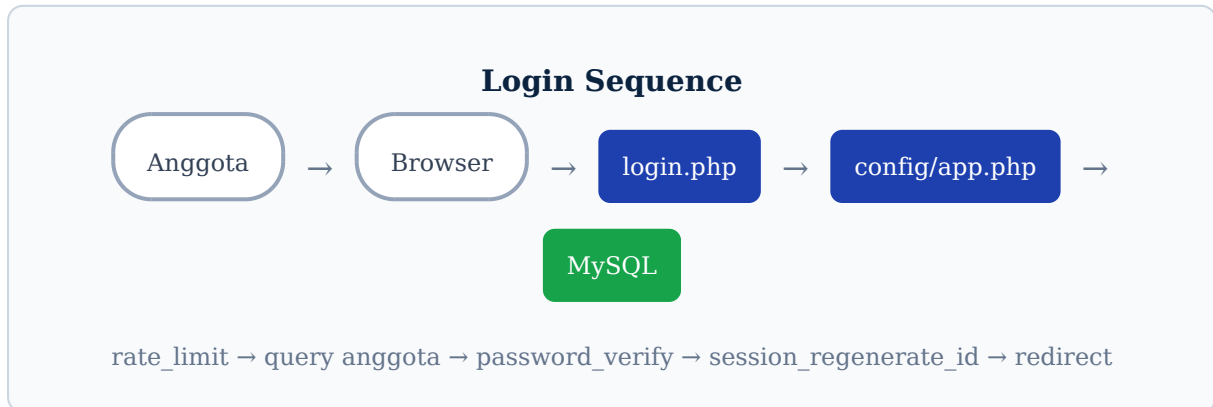
AKTIVITAS TERBARU	
Dipinjam: Laskar Pelangi (3 hari) Dikembalikan: Bumi (Rp 0) Rating: Bintang 5 untuk Bumi	
Riwayat Peminjaman Badge Pencapaian	

7.7 Badge Pencapaian

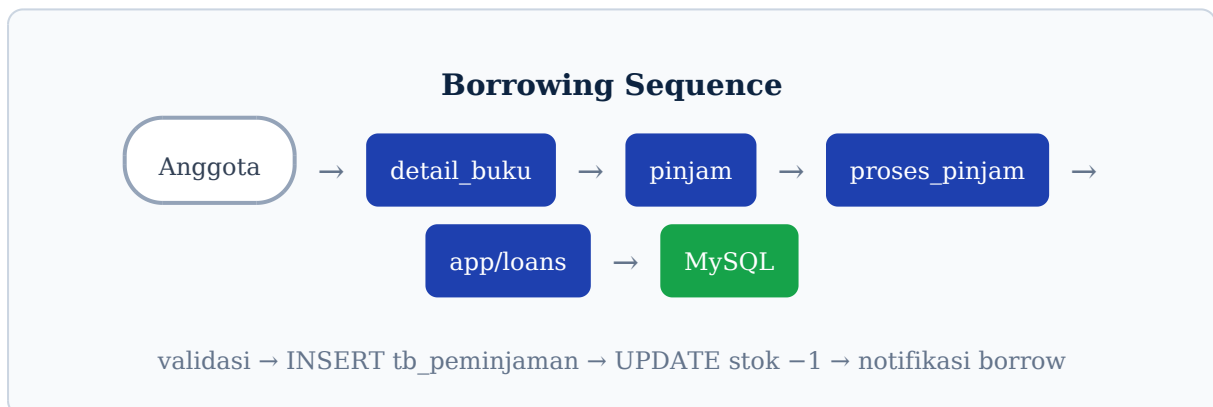
BADGE PENCAPAIAN (3 dari 3 badge diraih)		
Anggota Baru Teraih!	Pembaca Tekun Teraih!	Tepat Waktu Teraih!
PROGRESS SELANJUTNYA: Badge VIP: 3/10 pinjaman 30%		

8. Sequence Diagram

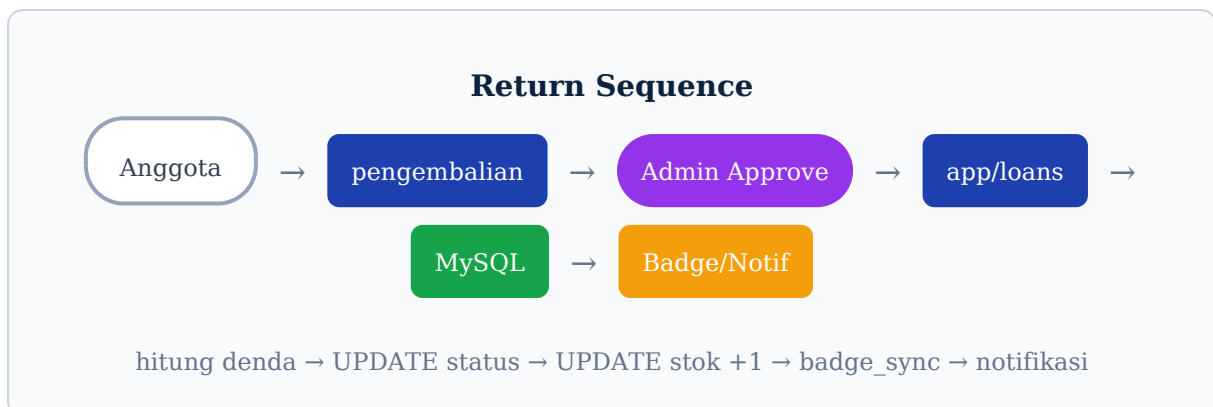
8.1 Sequence Login



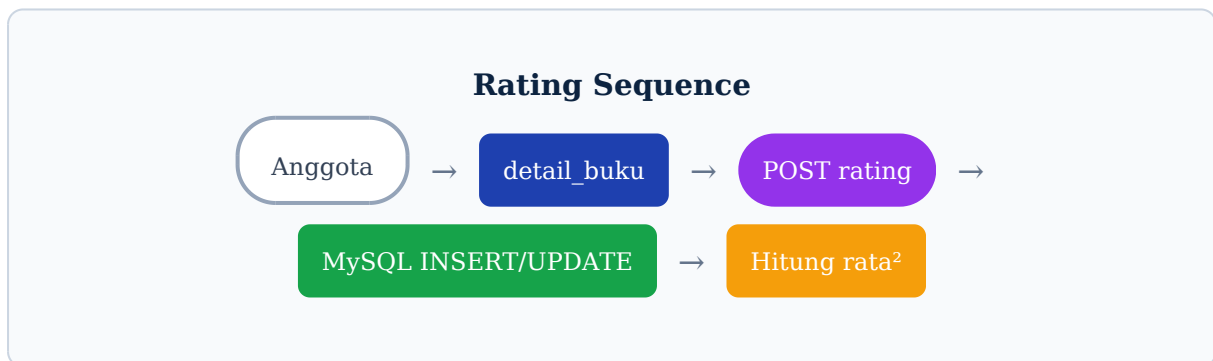
8.2 Sequence Peminjaman



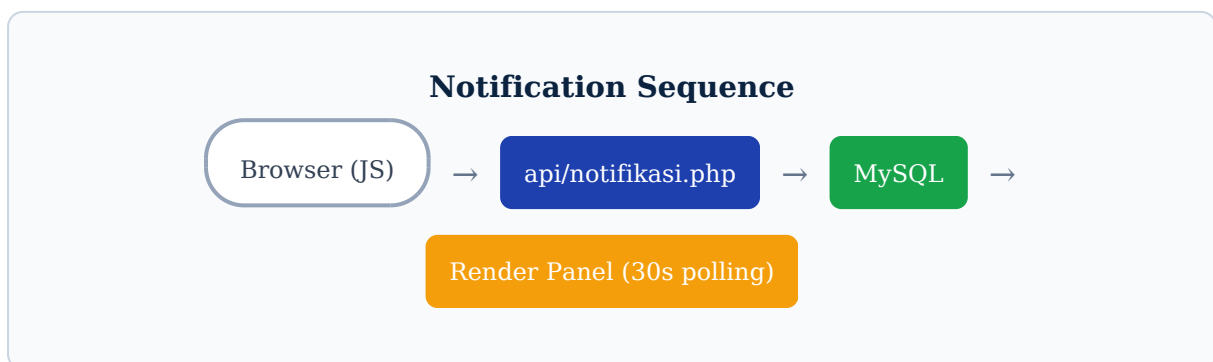
8.3 Sequence Pengembalian



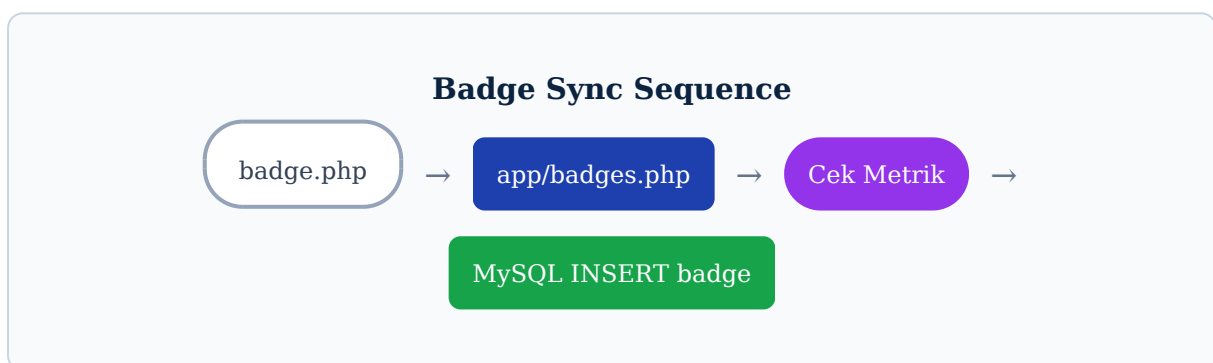
8.4 Sequence Rating



8.5 Sequence Notifikasi

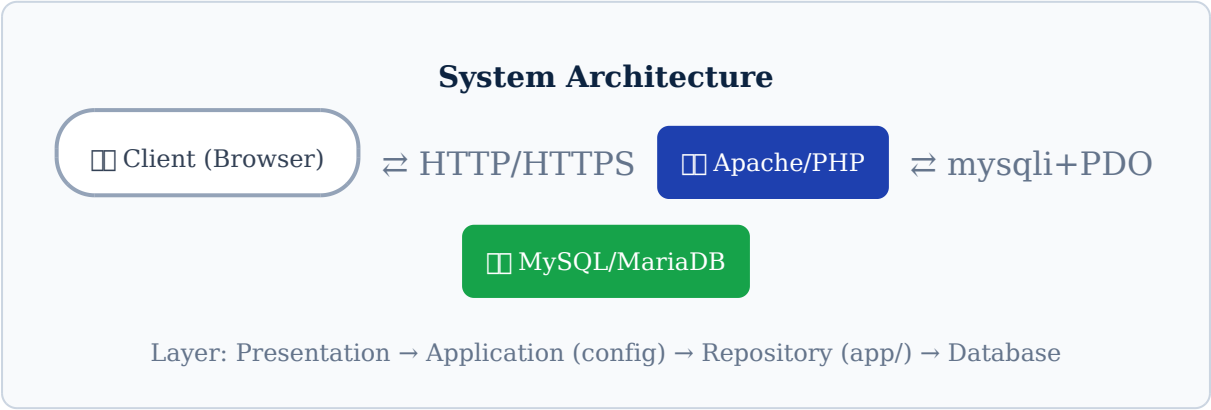


8.6 Sequence Badge Sync



9. Arsitektur Sistem

9.1 High-Level Architecture



9.2 Layer Architecture

Layer	Deskripsi	File
Presentation	UI templates, CSS, JS	*.php, assets/
Application	Business logic, auth, CSRF	config/app.php
Repository	Data access layer	app/*.php
Database	MySQL/MariaDB	config/database.php

9.3 Data Flow



9.4 Deployment Architecture



9.5 Technology Stack

Layer	Technology	Version	Purpose
Frontend	Bootstrap	5.3.2	UI Framework
Frontend	jQuery	3.7.1	DOM Manipulation
Frontend	SweetAlert2	11	Alerts
Frontend	DataTables	1.13.7	Table Management
Frontend	Font Awesome	6.5.1	Icons
Backend	PHP	8.0+	Server Logic
Database	MySQL/MariaDB	5.7+/10.3+	Data Storage
Security	Bcrypt	-	Password Hashing
PDF	dompdf	3.1	PDF Generation

10. Keamanan Sistem

10.1 Security Layers

Layer	Mekanisme	Implementasi
HTTP Headers	Security Headers	.htaccess
Authentication	Bcrypt + Session	config/app.php
Authorization	RBAC + Permission	tb_permission
CSRF	Token Rotation	config/app.php
XSS	Output Encoding	e() helper
SQL Injection	Prepared Statements	config/database.php
Rate Limiting	Session Counter	config/app.php
Session	Hardening	config/app.php

10.2 Password Hashing

```
password_hash($password, PASSWORD_DEFAULT) // Bcrypt
password_verify($input, $stored_hash)      // Verify
```

10.3 CSRF Protection

```
Token: bin2hex(random_bytes(32)) → 64 hex chars
Rotation: Setiap 30 menit
Verification: hash_equals() → timing-safe
```

10.4 Rate Limiting

Batas: 5 percobaan / 15 menit
Penyimpanan: \$_SESSION['login_attempts']
Efek: Akun terkunci sementara

10.5 HTTP Security Headers

X-Content-Type-Options: nosniff
X-Frame-Options: SAMEORIGIN
X-XSS-Protection: 1; mode=block
Referrer-Policy: strict-origin-when-cross-origin
Permissions-Policy: camera=(), microphone=()
Content-Security-Policy: (custom)

10.6 Security Checklist

No	Item	Status
1	Password Hashing (Bcrypt)	☐
2	CSRF Token	☐
3	XSS Protection (e())	☐
4	SQL Injection Prevention	☐
5	Rate Limiting	☐
6	Session Hardening	☐
7	Security Headers	☐
8	File Upload Validation	☐
9	RBAC	☐
10	IDOR Prevention	☐
11	Error Handling	☐
12	Honeypot Field	☐

11. Pengujian Sistem

11.1 Ringkasan Pengujian

Jenis Pengujian	Tool	Status
Unit Testing	PHPUnit	☐ Belum
Functional Testing	Manual	☐ Dilakukan
Security Testing	Manual	☐ Dilakukan
Responsiveness	DevTools	☐ Dilakukan
Cross-Browser	Multiple Browser	☐ Dilakukan

11.2 Test Cases: Autentikasi

No	Test Case	Expected	Status
TC-01	Login berhasil	Redirect dashboard	☐
TC-02	Login gagal - email salah	Error: email tidak ada	☐
TC-03	Login gagal - password salah	Error: password salah	☐
TC-04	Rate limiting 5x gagal	Akun terkunci 15 menit	☐
TC-05	Registrasi berhasil	Redirect login	☐
TC-06	Logout	Session dihapus	☐

11.3 Test Cases: Peminjaman

No	Test Case	Expected	Status
TC-07	Pinjam berhasil	Status dipinjam, stok -1	☐
TC-08	Stok habis	Error: stok habis	☐

TC-09	Duplikat pinjam	Error: sudah dipinjam	☐
TC-10	Batas tercapai	Error: batas tercapai	☐

11.4 Test Cases: Keamanan

No	Test Case	Expected	Status
TC-11	SQL Injection	Input ditolak	☐
TC-12	XSS	Output di-escape	☐
TC-13	CSRF tanpa token	Error 403	☐
TC-14	IDOR	Ditolak (session check)	☐

12. Deployment

12.1 Lingkungan

Aspek	Local (XAMPP)	Remote (InfinityFree)
Server	XAMPP	InfinityFree Hosting
PHP	8.0+	8.0+
Database	MySQL 5.7+	MariaDB 10.3+
URL	localhost/perpustakaan-daerah%20V5/	yourdomain.infinityfree.me/

12.2 Instalasi Local

1. Copy folder ke: C:
mmp\htdocs\perpustakaan-daerah V5\
2. Start Apache & MySQL (XAMPP)
3. Buat DB: phpmyadmin → perpustakaan_daerah
4. Import: database.sql
5. Buka: <http://localhost/perpustakaan-daerah%20V5/>
6. Login: admin / admin123

12.3 Instalasi Produksi

1. Buat akun InfinityFree
2. Buat hosting + database
3. Upload files ke htdocs/
4. Import database via phpMyAdmin
5. Edit config/database.php (kredensial hosting)
6. Edit config/app.php (APP_URL produksi)
7. Jalankan migrasi (jika perlu)

12.4 Post-Deployment Checklist

No	Item	Status
1	Database berhasil di-import	☐
2	Semua tabel terbuat (20)	☐
3	Login admin berhasil	☐
4	Login anggota berhasil	☐
5	Katalog buku tampil	☐
6	Peminjaman berfungsi	☐
7	Pengembalian berfungsi	☐
8	Rating berfungsi	☐
9	Notifikasi berfungsi	☐
10	Badge berfungsi	☐
11	CMS tampil	☐
12	Dark mode berfungsi	☐
13	Responsive mobile	☐
14	.htaccess aktif	☐

13. Database Schema

13.1 Daftar Lengkap Kolom

Tabel	Kolom	Tipe
tb_anggota	id_anggota	int(11) unsigned
	nomor_anggota	varchar(30)
	nama_lengkap	varchar(150)
	email	varchar(100)
	password	varchar(255)
	role	enum('anggota','admin')
	jenis_kelamin	enum('L','P')
	kelas	varchar(30)
	jurusan	varchar(50)
	alamat	text
	telepon	varchar(20)
	foto	varchar(255)
	status	enum('aktif','nonaktif')
	tanggal_daftar	date

	created_at	timestamp
	updated_at	timestamp
tb_badge	id_badge	int(11) unsigned
	kode	varchar(50)
	nama	varchar(100)
	deskripsi	varchar(255)
	ikon	varchar(50)
	warna	varchar(20)
	metrik	enum('total_pinjam','total_review','tepat_waktu')
	syarat	int(11) unsigned
	urutan	int(11)
	status	enum('aktif','nonaktif')
	created_at	timestamp
tb_badge_anggota	id_badge_anggota	int(11) unsigned
	id_anggota	int(11) unsigned
	id_badge	int(11) unsigned
	earned_at	timestamp
tb_banner	id_banner	int(11) unsigned

	judul	varchar(200)
	subjudul	varchar(255)
	gambar	varchar(255)
	tombol_text	varchar(100)
	tombol_url	varchar(255)
	urutan	int(11)
	status	enum('aktif','nonaktif')
	created_at	timestamp
	updated_at	timestamp
tb_berita	id_berita	int(11) unsigned
	judul	varchar(200)
	isi	text
	gambar	varchar(255)
	kategori	varchar(100)
	status	enum('aktif','nonaktif')
	created_at	timestamp
	updated_at	timestamp
tb_buku	id_buku	int(11) unsigned
	isbn	varchar(30)

judul_buku	varchar(255)
id_kategori	int(11) unsigned
id_penerbit	int(11) unsigned
id_rak	int(11) unsigned
pengarang	varchar(150)
tahun_terbit	year(4)
bahasa	varchar(50)
jumlah_halaman	int(11)
jumlah_buku	int(11)
stok_buku	int(11)
kondisi	enum('baik','rusak_ringan','rusak_berat')
foto	varchar(255)
sinopsis	text
keterangan	text
populer	tinyint(1)
baru	tinyint(1)
created_at	timestamp

	updated_at	timestamp
tb_faq	id_faq	int(11) unsigned
	pertanyaan	varchar(255)
	jawaban	text
	urutan	int(11)
	status	enum('aktif','nonaktif')
	created_at	timestamp
	updated_at	timestamp
tb_jam_buka	id_jam	int(11) unsigned
	hari	varchar(20)
	jam_buka	varchar(20)
	status	enum('buka','tutup')
	urutan	int(11)
tb_kategori	id_kategori	int(11) unsigned
	nama_kategori	varchar(100)
	keterangan	text
	created_at	timestamp
tb_layanan	id_layanan	int(11) unsigned
	nama	varchar(150)

	deskripsi	text
	ikon	varchar(50)
	urutan	int(11)
	status	enum('aktif','nonaktif')
	created_at	timestamp
	updated_at	timestamp
tb_notifikasi	id_notifikasi	int(11) unsigned
	id_target_user	int(11) unsigned
	id_target_staff	int(11) unsigned
	target_tipe	enum('admin','petugas','anggota')
	tipe	varchar(40)
	judul	varchar(200)
	pesan	text
	link	varchar(255)
	ref_table	varchar(50)
	ref_id	int(11) unsigned
	dibaca_at	datetime
	created_at	timestamp

tb_peminjaman	id_peminjaman	int(11) unsigned
	kode_peminjaman	varchar(30)
	id_anggota	int(11) unsigned
	id_buku	int(11) unsigned
	id_user	int(11) unsigned
	tanggal_pinjam	date
	tanggal_kembali	date
	tanggal_dikembalikan	date
	status	enum('dipinjam','terlambat','menunggu_pengembalian')
	denda	decimal(10,2)
	status_denda	enum('belum_dibayar','lunas')
	tanggal_diajukan	datetime
	tanggal_disetujui	datetime
	kondisi_saat_kembali	varchar(30)
	catatan_pengembalian	text
	diproses_oleh	int(11) unsigned
	keterangan	text

	created_at	timestamp
	updated_at	timestamp
tb_penerbit	id_penerbit	int(11) unsigned
	nama_penerbit	varchar(150)
	alamat	text
	telepon	varchar(20)
	email	varchar(100)
	created_at	timestamp
tb_pengaturan	id_pengaturan	int(11) unsigned
	nama_key	varchar(100)
	nilai	text
	tipe	enum('text','textarea','file')
	updated_at	timestamp
tb_pengumuman	id_pengumuman	int(11) unsigned
	judul	varchar(200)
	isi	text
	tipe	enum('info','peringatan','penting')
	status	enum('aktif','nonaktif')
	created_at	timestamp

	updated_at	timestamp
tb_permission	id_permission	int(11) unsigned
	role	enum('admin','petugas','anggota')
	modul	varchar(50)
	akses	enum('baca','tulis','semua')
tb_rak	id_rak	int(11) unsigned
	nama_rak	varchar(50)
	lokasi	varchar(100)
	keterangan	text
	created_at	timestamp
tb_rating	id_rating	int(11) unsigned
	id_buku	int(11) unsigned
	id_anggota	int(11) unsigned
	rating	tinyint(1)
	komentar	text
	status	enum('tampil','disembunyikan')
	created_at	timestamp
	updated_at	timestamp

tb_sesi_login	id_sesi	int(11) unsigned
	id_anggota	int(11) unsigned
	token_hash	char(64)
	expires_at	datetime
	created_at	timestamp
tb_user	id_user	int(11) unsigned
	nama_user	varchar(100)
	username	varchar(50)
	password	varchar(255)
	level	enum('admin','petugas')
	foto	varchar(255)
	status	enum('aktif','nonaktif')
	created_at	timestamp
	updated_at	timestamp

13.2 Stored Procedures

Procedure	Deskripsi	File
sp_pinjam_buku	Proses peminjaman buku	migrasi_prosedur.sql

sp_kembalikan_buku	Proses pengembalian buku	migrasi_prosedur.sql
sp_hitung_denda	Kalkulasi denda keterlambatan	migrasi_prosedur.sql

13.3 Triggers

Trigger	Event	Deskripsi
trg_buku_before_insert	BEFORE INSERT	Validasi stok $\leq$ jumlah
trg_buku_before_update	BEFORE UPDATE	Validasi stok $\leq$ jumlah
trg_peminjaman_after_insert	AFTER INSERT	Update stok -1

13.4 Functions

Function	Return	Deskripsi
fn_generate_kode_pinjam	VARCHAR(20)	Generate kode PMJ-YYYYMMDD-XXX
fn_hitung_denda	INT	Kalkulasi denda dari selisih hari

14. API Reference

14.1 Notifikasi API

Endpoint: `api/notifikasi.php`

GET ?aksi=list

Parameter	Tipe	Default	Deskripsi
aksi	string	-	"list"
limit	int	8	Jumlah notifikasi

Response:

```
{
  "unread": 3,
  "items": [
    {
      "id_notifikasi": 1,
      "tipe": "borrowing",
      "judul": "Peminjaman Baru",
      "pesan": "...",
      "dibaca_at": null,
      "created_at": "2026-09-02 10:00:00"
    }
  ]
}
```

POST ?aksi=mark_read

Parameter	Tipe	Deskripsi
aksi	string	"mark_read"
id	int	ID notifikasi (opsional, semua jika kosong)

Response:

```
{
  "success": true,
  "unread": 2
}
```

14.2 Flash Messages API

Gunakan `flash()` helper untuk pesan sesi:

```
flash('success', 'Buku berhasil dipinjam!');
// → Set $_SESSION['flash'] dengan SweetAlert2 config
// → Ditampilkan oleh header.php setelah redirect
```

14.3 CSRF Token API

```
// Di form HTML:
<input type="hidden" name="csrf_token" value="= csrf_token() ?">

// Di JavaScript:
headers: { 'X-CSRF-TOKEN': '= csrf_token() ?' }

// Di server:
csrf_verify(); // Cek token
```

15. File Structure

15.1 Struktur Direktori

```
perpustakaan-daerah V5/
├── .htaccess                # Apache security headers + URL rewrite
├── composer.json            # PHP dependencies (dompdf)
├── composer.lock            # Locked dependencies
├── database.sql              # Main DB schema (20 tables)
├── migrasi_v5.sql            # V5 migration
├── migrasi_badge_informasi.sql # Badge migration
├── migrasi_notifikasi.sql    # Notification migration
├── migrasi_rating_komentar.sql # Rating migration
├── migrasi_prosedur.sql       # Stored procedures
├── generate_pdf.php          # PDF generator script
├──
├── config/
│   ├── app.php              # App config, helpers, auth, CSRF
│   └── database.php          # DB connection (mysqli + PDO)
├──
├── app/
│   ├── catalog.php          # Book catalog repository
│   ├── loans.php            # Loan repository
│   ├── ratings.php           # Rating & comment repository
│   ├── notifications.php     # Notification system
│   ├── badges.php            # Badge/achievement system
│   ├── anggota.php           # Member repository
│   └── content.php           # CMS content repository
├──
├── api/
│   └── notifikasi.php        # Notification AJAX endpoint
├──
├── partials/
│   ├── header.php           # HTML header + nav + drawer
│   ├── footer.php           # HTML footer
│   ├── drawer.php            # Mobile drawer navigation
│   ├── book_card.php         # Book card component
│   ├── landing_book_card.php # Landing page book card
│   └── library_video.php     # Library video component
```

```

|   └─ notifications.php      # Notification bell component
|
|   └─ assets/
|       └─ css/
|           └─ design-tokens.css  # CSS custom properties
|           └─ v5.css            # V5 custom styles
|           └─ enterprise.css     # Enterprise theme
|           └─ style.css         # Main overrides
|           └─ components/       # UI component styles
|           └─ layout/          # Layout styles
|           └─ pages/           # Page-specific styles
|               └─ themes/      # Theme variants
|
|       └─ js/
|           └─ app.js            # Core JS (helpers, auth, CSRF)
|           └─ ui/
|               └─ drawer.js      # Mobile drawer
|               └─ notifications.js # Notification system
|               └─ theme.js       # Theme switcher
|               └─ toast.js       # Toast notifications
|               └─ pages/
|                   └─ landing.js  # Landing page JS
|                   └─ library_video.js # Video player JS
|
|       └─ img/
|           └─ logo.svg          # Library logo
|           └─ default-cover.svg # Default book cover
|           └─ empty-state.svg   # Empty state illustration
|           └─ illustrations/    # Various illustrations
|
|   └─ uploads/
|       └─ buku/                # Book covers
|       └─ user/                # User profile photos
|       └─ .htaccess            # Disable PHP execution
|
|   └─ docs/                    # Documentation (this folder)
|       └─ 01_erd.md
|       └─ 02_use_case_diagram.md
|       └─ 03_user_flow.md
|       └─ 04_admin_flow.md
|       └─ 05_flowchart.md
|       └─ 06_algoritma.md
|       └─ 07_mockup_ui.md

```

```

|   └─ 08_sequence_diagram.md
|   └─ 09_architecture_diagram.md
|   └─ 10_keamanan_sistem.md
|   └─ 11_pengujian_sistem.md
|   └─ 12_deployment.md
|
└─ index.php           # Landing page
└─ login.php           # Login
└─ register.php        # Registration
└─ logout.php          # Logout
└─ katalog_buku.php    # Book catalog
└─ detail_buku.php     # Book detail
└─ cari.php            # Search
└─ pinjam.php          # Loan form
└─ proses_pinjam.php   # Loan processor
└─ pengembalian.php    # Return management
└─ riwayat_peminjaman.php # Loan history
└─ akun.php            # Member dashboard
└─ badge.php           # Badge achievements
└─ informasi.php       # CMS information
└─ help.php            # Help page

```

15.2 File Count Summary

Category	Count	Description
PHP Pages	16	Main application pages
Config	2	Configuration files
App (Repository)	7	Data access layer
API	1	JSON endpoints
Partials	7	Reusable components
CSS	12+	Stylesheets
JavaScript	8	Scripts
SQL	6	Database scripts
Documentation	12	docs/ folder

Total

70+

16. Design System

16.1 CSS Architecture

Layer	File	Deskripsi
1	design-tokens.css	CSS custom properties (warna, font, spacing)
2	style.css	Main overrides & utilities
3	v5.css	V5 custom component styles
4	enterprise.css	Enterprise theme variant
5	components/	UI component styles
6	layout/	Layout & grid styles
7	pages/	Page-specific styles
8	themes/	Theme variants

16.2 Color Tokens

Token	Light Mode	Dark Mode	Usage
--color-primary	#1e40af	#3b82f6	Primary actions
--color-primary-soft	#dbeafe	#1e3a5f	Light backgrounds
--color-success	#16a34a	#4ade80	Success states
--color-warning	#f59e0b	#fbbf24	Warning states
--color-danger	#dc2626	#f87171	Error states
--color-bg	ffffff	#0f172a	Page background
--color-text	#1a1a1a	#e2e8f0	Main text

16.3 Typography

Element	Font	Size	Weight
Body	Inter	14px	400
H1	Inter	24px	700
H2	Inter	20px	600
H3	Inter	16px	600
Small	Inter	12px	400

16.4 JavaScript Architecture

File	Purpose
app.js	Core: helpers, auth, CSRF, flash messages
ui/drawer.js	Mobile drawer navigation
ui/notifications.js	Notification polling & rendering
ui/theme.js	Theme switcher (light/dark)
ui/toast.js	Toast notification system
pages/landing.js	Landing page interactions
pages/library_video.js	Video player controls

16.5 Theme Support

- **Light Mode** — Default theme
- **Dark Mode** — Toggle via header icon
- **Persistence** — Saved in localStorage
- **Implementation** — CSS custom properties on `data-theme` attribute

17. Configuration

17.1 Environment Config (config/database.php)

```
define('DB_ENV', 'local'); // 'local' | 'infinityfree'

// Local (XAMPP)
define('DB_HOST', 'localhost');
define('DB_NAME', 'perpustakaan_daerah');
define('DB_USER', 'root');
define('DB_PASS', '');

// InfinityFree
define('DB_HOST', 'sqlXXX.infinityfree.com');
define('DB_NAME', 'if0_XXXXXX_nama_db');
define('DB_USER', 'if0_XXXXXX');
define('DB_PASS', 'password');
```

17.2 Application Config (config/app.php)

```
define('APP_URL', 'http://localhost/perpustakaan-daerah%20V5');
define('BASE_URL', APP_URL . '/');
define('APP_NAME', 'Perpustakaan Daerah');
define('APP_VERSION', '5.0');
define('APP_DEBUG', false);
define('SESSION_TIMEOUT', 1800); // 30 menit
define('MAX_PEMINJAMAN', 2);
define('DENDA_PER_HARI', 10000);
define('LOAN_DURATION_DAYS', 7);
define('RATE_LIMIT_MAX', 5);
define('RATE_LIMIT_TIME', 900); // 15 menit
```

17.3 .htaccess Config

```
# Security Headers
Header set X-Content-Type-Options "nosniff"
Header set X-Frame-Options "SAMEORIGIN"
Header set X-XSS-Protection "1; mode=block"
Header set Referrer-Policy "strict-origin-when-cross-origin"
Header set Permissions-Policy "camera=(), microphone=(), geolocation=()"
Header set Content-Security-Policy "...

# File Protection
<FilesMatch \"\.(env|git|htaccess|htpasswd)$\">
    Require all denied
</FilesMatch>

# Upload Protection
<Directory \"uploads\">
    <FilesMatch \"\.(php|phtml|php3|php4|php5)$\">
        Require all denied
    </FilesMatch>
</Directory>
```

17.4 Constants Summary

Constant	Value	File
APP_URL	http://localhost/perpustakaan-daerah%20V5	app.php
APP_VERSION	5.0	app.php
SESSION_TIMEOUT	1800 (30 min)	app.php
MAX_PEMINJAMAN	2	app.php
DENDA_PER_HARI	10000 (Rp)	app.php
LOAN_DURATION_DAYS	7	app.php
RATE_LIMIT_MAX	5	app.php
RATE_LIMIT_TIME	900 (15 min)	app.php

18. Changelog & Credits

18.1 Changelog

V5.0 (2 September 2026)

- Redesign total dengan Bootstrap 5.3.2
- Implementasi Dark Mode dengan CSS Custom Properties
- Sistem Notifikasi Real-Time (AJAX polling 30 detik)
- Sistem Rating & Komentar Buku
- Sistem Badge Pencapaian (3 metrik)
- CMS: Berita, Pengumuman, Banner, Layanan, FAQ, Jam Buka
- 20 Database Tables
- Prepared Statements (mysqli + PDO)
- Rate Limiting (5 attempt / 15 min)
- CSRF Token Rotation (30 min)
- Persistent Login (Remember Me)
- Mobile-First Responsive Design
- Component-Based UI (partials/)
- Design Token System (CSS Custom Properties)
- Enterprise Theme Support
- Google Fonts (Inter + Poppins)
- Font Awesome 6.5.1 Icons
- SweetAlert2 for Alerts
- DataTables for Tables
- dompdf for PDF Generation

V4.0 (Sebelumnya)

- Bootstrap 5 + AdminLTE

- PHP 7+ Native
- MySQL Database
- CRUD Operations
- User Authentication

18.2 Credits

Technology	Author	License
Bootstrap 5.3.2	Bootstrap Team	MIT
jQuery 3.7.1	jQuery Foundation	MIT
Font Awesome 6.5.1	Fonticons	MIT + CC BY 4.0
SweetAlert2 11	Limonte	MIT
DataTables 1.13.7	Allan Jardine	MIT
Inter Font	Rasmus Andersson	OFL 1.1
Poppins Font	Indian Type Foundry	OFL 1.1
dompdf 3.1	dompdf Team	MIT
PHP 8.0+	PHP Group	PHP License
MySQL/MariaDB	Oracle / MariaDB Foundation	GPL

18.3 Dokumen Terkait

Dokumen	Lokasi
ERD	docs/01_erd.md
Use Case Diagram	docs/02_use_case_diagram.md
User Flow	docs/03_user_flow.md
Admin Flow	docs/04_admin_flow.md

Flowchart	docs/05_flowchart.md
Algoritma	docs/06_algoritma.md
Mockup UI	docs/07_mockup_ui.md
Sequence Diagram	docs/08_sequence_diagram.md
Arsitektur	docs/09_architecture_diagram.md
Keamanan	docs/10_keamanan_sistem.md
Pengujian	docs/11_pengujian_sistem.md
Deployment	docs/12_deployment.md
PDF Dokumentasi	DOKUMENTASI_PERPUSTAKAAN_DAERAH_V5.pdf
Project Planning	PERENCANAAN_PROYEK_PERPUSTAKAAN_DAERAH_V5.md

DOKUMENTASI SISTEM INFORMASI PERPUSTAKAAN DAERAH VERSI 1

Digenerate pada: "" . PDF_DATE . ""

Tim Pengembang Sistem Perpustakaan Daerah (Versi 1)

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